

$\exists$ is the Image

by notes on Freyd & Scedrov, Categories · Allegories

The existential quantifier as direct image — §1.451 (inverse image), §1.6 (pre-logos)

Fix a map $f : A \rightarrow B$. It carries subsets **forward** (direct image $\exists_f$) and **backward** (inverse image $f^\#$); the forward one **is** the existential quantifier. We watch this in SET element by element, see why projection gives $\exists y$, and read off the adjunction $\exists_f \models f^\#$.

Remark 0.1 (Setup).

$f : A \rightarrow B$ a map. A **subset** $S \subseteq A$ is a unary predicate on A . Composition is written in **diagram order**: $S \hookrightarrow A \xrightarrow{f} B$ means “first include S , then apply f ”. In SET the operative element rule is: apply f to each element.

§1. Direct image, element by element

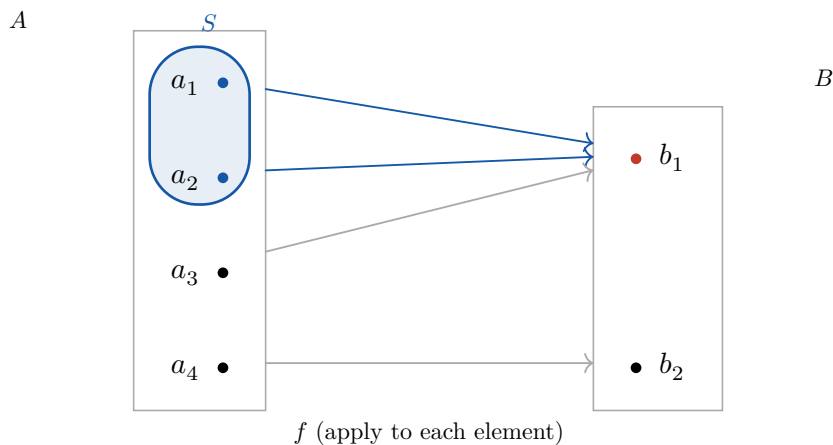
Definition 1.1 (Direct image $\exists_f$).

For $S \subseteq A$,

$$\exists_f(S) = f(S) = \{b \in B \mid \exists a \in S. f(a) = b\}.$$

The “ $\exists a$ ” is **literally in the definition**: one element of S landing on b is enough.

b lies in $\exists_f(S)$ iff its **fibre** $f^{-1}(b)$ **meets** S . Below, $S = \{a_1, a_2\}$ is the blue circle.

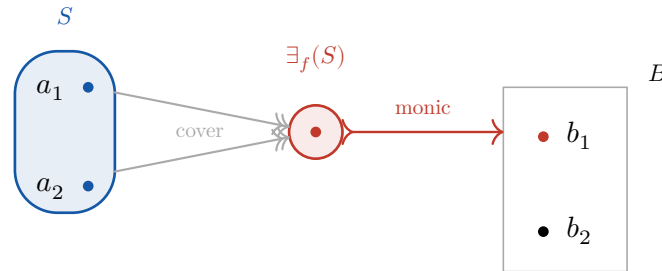


★ Punchline

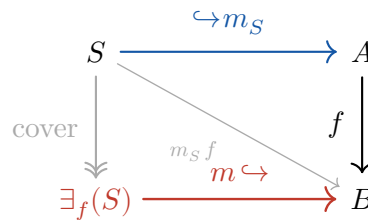
Meets, not contains. The fibre over b_1 is $\{a_1, a_2, a_3\}$ — it only **overlaps** S (the witness $a_3 \notin S$), yet $b_1 \in \exists_f(S)$: one witness in S is enough. The fibre over b_2 is $\{a_4\}$ and misses S entirely, so $b_2 \notin \exists_f(S)$. Requiring the **whole** fibre $\subseteq S$ would instead give the universal image $\forall_f(S)$ — the $\forall$ quantifier of the last section, a strictly different, dual construction.

§2. The categorical picture: image factorization

In $\mathbf{SET}$ the composite $S \hookrightarrow A \xrightarrow{f} B$ splits in two — collapse equal values, then include. With $S = \{a_1, a_2\}$, both sent to b_1 :



The **cover** fuses a_1, a_2 onto one point; the **monic** includes that point into B as the subobject $\exists_f(S)$. A general category has no elements, but this same factorization of $S \hookrightarrow A \xrightarrow{f} B$ — cover then monic — **is** the definition:



Definition 2.1 (In the repo (Freyd/S1_60.lean)).

The diagram's monic is $m_S = S.\text{arr}$ (the inclusion $S \hookrightarrow A$), so

$$\exists_f(S) := \text{image}(m_S f),$$

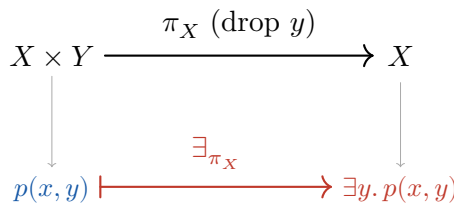
the subobject $\exists_f(S) \hookrightarrow B$ drawn above. In Lean: `existsAlong f S := image (S.arr >> f)` (generic signature `existsAlong g U`). The adjunction (next section) is `existsAlong_le_iff`. Needs only a **regular** category, not the full pre-logos.

§3. Why projection gives $\exists y$

Take $f = \pi_X : X \times Y \rightarrow X$, the projection that **forgets** Y . A subset $P \subseteq X \times Y$ is a **two-place** predicate $p(x, y)$. Push it forward:

$$\exists_{\pi_X}(P) = \{x \in X \mid \exists(x', y) \in P. x' = x\} = \{x \mid \exists y. p(x, y)\}.$$

So $\exists_{\pi_X}$ along the projection that drops Y **is** the quantifier $\exists y$ — the source of the name.

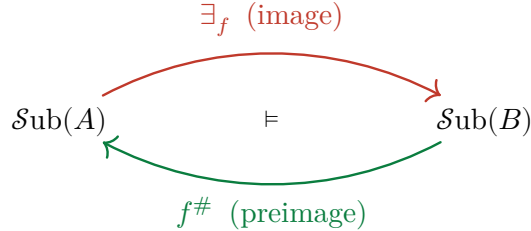


§4. The adjunction $\exists_f \models f^\#$

Forward image is **left adjoint** to backward image — a **Galois connection** between the subset lattices:

$$\exists_f(S) \subseteq T \iff S \subseteq f^{\#(T)} \quad (S \subseteq A, T \subseteq B).$$

Both sides say “ S maps into T ”; in SET, $\text{all } (\in T) \text{ (map } f \text{ } S) = \text{all } (\lambda a. f \ a \in T) \ S$.



Remark 4.1 (The whole hierarchy in one line).

$f^{\#}$ is the substitution in the middle of $\exists_f \models f^{\#} \models \forall_f$.

- $\exists_f \models f^{\#}$ — a **regular** category. Gives $\exists$ (and that $f^{\#}$ preserves $\cap$, “for free”, as a right adjoint).
- $f^{\#} \models \forall_f$ — a **Heyting** category (Freyd’s **logos**). Gives $\forall$.

The **pre-logos** (coherent) layer in between is exactly: $f^{\#}$ **also** preserves unions $\cup$ — the one preservation that is **not** free, and the content behind the §1.616 proofs in `S1_60.lean`.

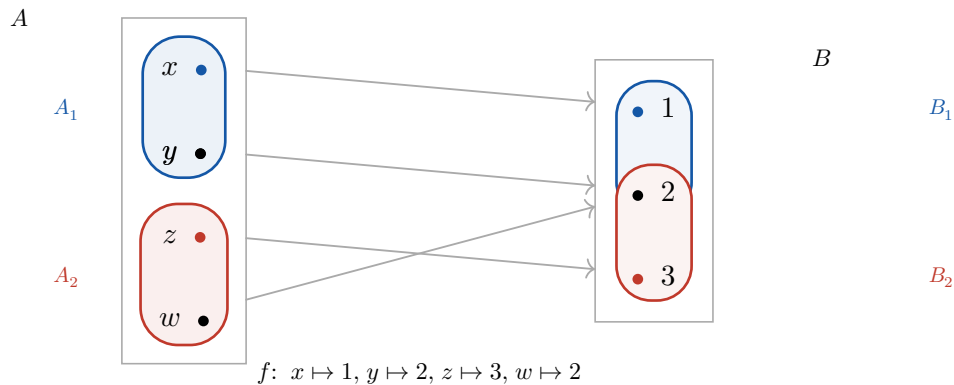
§5. The third definition: “pullbacks transfer finite covers”

Freyd gives a **third**, equivalent definition of pre-logos (§1.611) that never mentions lattices:

A **Cartesian category with images** in which **pullbacks transfer finite covers**: given a cover $\{B_i \twoheadrightarrow B\}$ and any map $A \rightarrow B$, the pullbacks $A_i = A \times_B B_i$ form a cover $\{A_i \twoheadrightarrow A\}$.

Unpacked in SET: a **cover** is a surjection, and a finite family $\{B_i \twoheadrightarrow B\}$ is a cover when it is **jointly surjective** — every $b \in B$ is hit by some B_i , i.e. as subobjects $\cup_i B_i = B$. Pulling back along f is taking the **preimage**. The claim: preimage of a jointly-surjective family is jointly surjective.

§5.1. A concrete instance



The cover of B is $B_1 = \{1, 2\}$, $B_2 = \{2, 3\}$ (jointly $= B$). Their preimages

$$A_1 = f^{\#(B_1)} = \{x, y, w\}, \quad A_2 = f^{\#(B_2)} = \{y, z, w\}$$

satisfy $A_1 \cup A_2 = \{x, y, z, w\} = A$ — so $\{A_1, A_2\}$ covers A . The cover **transferred down the pullback**. (Note y, w land in **both** A_i because $2 \in B_1 \cap B_2$; overlap is fine.)

§5.2. Why this is the same axiom, covering-side-up

A cover is the equation $B_1 \cup B_2 = B$ (the union is the top $\top_B$). “Transfers” then says

$$f^{\#(B_1 \cup B_2)} = f^{\#(B_1)} \cup f^{\#(B_2)} \quad (= \top_A),$$

which is just $f^{\#}$ **preserving unions** — Definition 2 — in the special case where the union is the whole object. Relativizing $\top$ to an arbitrary subobject recovers the general statement. So Definitions 2 and 3 are the **same** condition, stated bottom-up (subobjects) vs. top-down (covers).

★ Punchline

In **SET** it is automatic; in general it is a real axiom. The one-line proof: for $a \in A$, $f(a)$ is hit by some B_i , so $a \in f^{\#(B_i)}$ — preimage of jointly-surjective is jointly-surjective. This works because **SET** has **extensive coproducts** (f^{-1} commutes with disjoint union), i.e. it is **positive** (§1.623). In a bare regular category the pullback squares always **exist**, but the forward inclusion $f^{\#(B_1 \cup B_2)} \leq f^{\#(B_1)} \cup f^{\#(B_2)}$ can fail — which is exactly why Freyd states “transfer finite covers” as a **hypothesis**, and exactly the extensivity wall behind the sorries in `S1_60.lean`.

§6. §1.611: the three sentences that define a pre-logos

§1.611 packages the whole definition as “a **Cartesian category with images such that**” exactly three geometric sentences hold. Each has the shape $\forall(\text{given}) \dots \exists(\text{built}) \dots$, and each is one of the new fields **PreLogos** adds over a regular category (`bottom`, `union`, `invImage_preserves_union slash _bottom` — rendered by `Fredy/tool/QSeqRender.lean`):

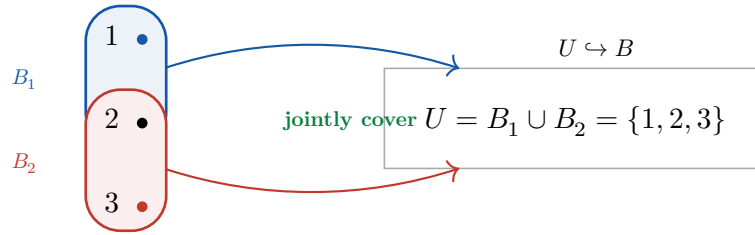
- (i) **Empty union** $\rightarrow$ the minimal subobject 0 (logic: $\perp$)
- (ii) **Binary union** $\rightarrow$ any two subobjects have a join $B_1 \cup B_2$ (logic: $\vee$)
- (iii) **Cover-transfer** $\rightarrow f^{\#}$ preserves those unions (logic: substitution commutes with $\perp, \vee$)

§6.1. (i) Empty union $= \perp$ — the minimal subobject

$\exists A, \forall(f : B \rightarrow A), \exists g$ inverse to f . The join of **zero** subobjects is the bottom 0 ; “every morphism to it is an isomorphism” makes it the strict initial object (§1.61). In **SET**, $0 = \emptyset$, and $\mathcal{S}ub(0) = \{0\}$ — no proper subobject.

§6.2. (ii) Binary union — the join of two subobjects

$\forall$ subobjects $B_1, B_2 \hookrightarrow B$, $\exists U$ with B_1, B_2 **jointly covering** U , and $U \hookrightarrow B$ monic. (§1.513 **covering pair**: drawn as one arc — no proper subobject of U allows **both**; neither B_i alone need cover U .)



In **SET** with $B = \{1, 2, 3\}$, $B_1 = \{1, 2\}$, $B_2 = \{2, 3\}$: $U = \{1, 2, 3\}$, and the two inclusions are **jointly** surjective onto U (2 hit by both — overlap fine). This is the $\vee$ of $\mathcal{S}ub(B)$.

§6.3. (iii) Cover-transfer — $f^\#$ preserves the join

$\forall$ a cover $\{B_1, B_2\}$ of B and a map $A \rightarrow B$, $\exists$ pullbacks $\{A_1, A_2\}$ jointly covering A — the worked example of the previous section. It says inverse image commutes with $\vee$ (and with $\perp$): $f^\#(B_1 \cup B_2) = f^\#(B_1) \cup f^\#(B_2)$.

§6.4. How the three fit together

have	get
Cartesian + images	$\cap, \top, \exists$ (and $f^\#$ preserves $\cap$, free)
+ (i) empty union	$\perp = 0$ — $\mathcal{S}ub(B)$ acquires a bottom
+ (ii) binary union	$\vee$ — $\mathcal{S}ub(B)$ is now a full lattice
+ (iii) cover-transfer	$f^\#$ preserves $\perp, \vee$ — each $f^\#$ a lattice hom
= pre-logos	the §1.6 definition, in elementary form

★ Punchline

The three sentences are **exactly** the gap between regular and pre-logos: (i)+(ii) make each $\mathcal{S}ub(B)$ a lattice (not just a semilattice), and (iii) makes every $f^\#$ a lattice homomorphism. So §1.611's three Q-sentences **are** the elementary restatement of the §1.6 definition — the lattice/lattice-hom phrasing and the cover phrasing are the same axioms.

§7. §1.612: a pre-logos forces every $\mathcal{S}ub(B)$ distributive

For a **monic** $f : A \hookrightarrow B$ the inverse image is intersection with that subobject, $f^\#(B') = A \cap B'$ (§1.612). Freyd's observation:

$f^\#$ preserves binary unions for **all** monic f targeted at $B \iff \mathcal{S}ub(B)$ is a **distributive lattice**: $A \cap (B_1 \cup B_2) = (A \cap B_1) \cup (A \cap B_2)$.

In a pre-logos $f^\#$ preserves unions for **every** f — in particular every monic — so this is automatic:

$$\text{pre-logos} \implies \mathcal{S}ub(B) \text{ distributive, for every } B.$$

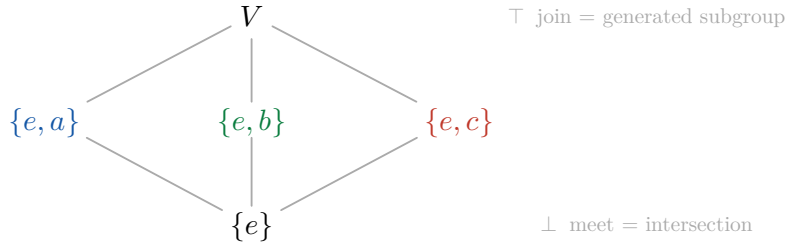
You cannot have a pre-logos with a non-distributive subobject lattice.

§7.1. In SET: always distributive

$\mathcal{S}ub(B) = \mathcal{P}(B)$, the full powerset — a **Boolean algebra**, hence distributive. Meet is $\cap$, join is $\cup$, and the law is the ordinary set identity. So for every B , $\mathcal{S}ub(B)$ in SET is distributive. (SET is a pre-logos, consistent with the implication above.)

§7.2. A non-distributive $\mathcal{S}ub(B)$: groups, the Klein four-group

Outside a pre-logos this can fail. In GRP take $B = V = \{e, a, b, c\}$, the Klein four-group. Its subobjects (= subgroups) form $\mathbf{M}_3$, the “diamond” — three incomparable atoms:



Test distributivity on the atom $\{e, a\}$:

$$\{e, a\} \cap (\{e, b\} \cup \{e, c\}) = \{e, a\} \cap V = \{e, a\},$$

$$(\{e, a\} \cap \{e, b\}) \cup (\{e, a\} \cap \{e, c\}) = \{e\} \cup \{e\} = \{e\}.$$

Since $\{e, a\} \neq \{e\}$, $\mathbf{M}_3$ is **not distributive**. The culprit: in GRP the join is the **generated** subgroup, so $\{e, b\} \cup \{e, c\} = V$ overshoots the set-union $\{e, b, c\}$ (not even a subgroup). GRP is therefore **not** a pre-logos — exactly as §1.612 predicts.

★ Punchline

“Can $\mathcal{S}ub(B)$ fail to be distributive?” — **yes in general** (GRP, $\mathbf{M}_3$), but **never inside a pre-logos**, and never in SET. §1.612 is the reason: the pre-logos axiom (union preserved under inverse image), specialized to monics, **is** the distributive law.